

TAMPINES JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
[mark scheme]



CANDIDATE
NAME

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CIVICS GROUP

1	2			
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TUTOR
NAME

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Biology

9648/03

Applications Paper and Planning Question

Friday, 20 September 2013

Paper 3

2 hours

Candidates answer on the Question Paper

Additional Materials are required: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination:

1. Fasten all answer scripts for Question 5 securely;
2. Hand in Paper 3 and the answer scripts for
Question 5 **separately**.

For Examiner's Use		
1	15	
2	15	
3	10	
4	12	
5	20	
Total	72	

The number of marks is given in brackets [] at the end of each question or part question.

STRUCTURED QUESTIONS (40 MARKS)

Answer **all** the questions in this section in the space provided.

QUESTION ONE (15 MARKS)

One of the applications of RFLP analysis is for restriction mapping. *Bam*H1, *Eco*RI and *Hind*III restriction enzymes are used to map a 15kb bacterial DNA. Three reactions were set up to cut the 15kb DNA sample, each with one of the three enzymes. After digestion with restriction enzymes, the DNA fragments were separated into bands. Results of the reactions are shown in Fig 1.1.

- (b) Describe briefly how, in such DNA analysis,
- (i) DNA is separated into bands; [4]
- **via gel electrophoresis;**
 - **negatively charged DNA migrate towards positive electrode / anode;**
 - **DNA fragments thus separated by their molecular size / length ;**
 - **smaller / shorter restriction fragments migrate faster / move a longer distance in a fixed time;**

- (ii) the bands are made visible. [1]
- **DNA fragments are stained with Et Br**
 - **which intercalate between DNA**
 - **visualize under UV light**

Or

- **stain with methylene blue solution**
- **view with naked eyes**

- (iii) Based on the information given and the results in *Fig 1.1*, which of the following map (W, X, Y or Z) in *Fig 1.2* is not a possible map? [1] **A**

- (iv) With reference to the results shown in Fig 1.3, explain which of the following W, X Y or Z is the correct map. [3]

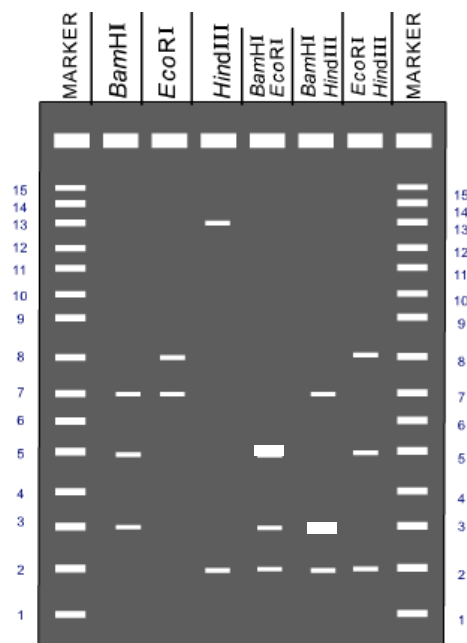


Fig 1.3

- **Y;;**

- When the 15kb DNA is cut with the two enzyme BamHI and EcoRI, there are three bands, 5kb, 3kb and 2kb
- the 5kb band is thicker- two copies of the 5kb
- when cut with the BamHI and HindIII, there are 3 bands, 7kb, 3kb and 2kb,
- where 3kb is the doublet
- When cut with enzyme EcoRI and HindIII, there are 3 bands, 8kb, 5kb and 2kb

(c) Another application of RFLP analysis is for forensic scientist to identify criminals. Two men (Suspect 1 and 2) were accused of kidnapping and raping a woman. RFLP analysis was used to examine the genetic characteristics of the two suspects, the victim, and a vaginal aspirate. Two genetic loci, D4S139 (a) and D10S28 (b), were examined. These two loci are variable number of tandem repeats (VNTR).

Fig 1.4 shows two autoradiograms of the DNA banding patterns. Three lanes of the autoradiograms (on the far left, far right, and middle) display multiple bands called size markers.

(i) State the function of the size markers. [2]

- **Size markers are DNA fragments of known sizes;**
- **They are compared to the bands in the other lanes;**
- **in order to allow the size of these fragments to be estimated;**

1max

(ii) Explain the term VNTR. [1]

- **non coding DNA portions containing repeated sequences of between 20 and 100 base pairs;**
- **sequence repeat in series/one after another;**

(iii) Identify the criminal with evidences shown in Fig 1.4. [3]

- **Suspect 2;**
- **when genetic locus D4S139 were used, banding pattern of suspect two (lane 4) is exactly the same as the semen from vagina (lane 7) in (a);**
- **when genetic locus D10S28 were used, one DNA band (lowest band) in suspect 1's matches the lowest band of the semen from the vagina,**
- **however, there are two DNA bands from the suspect one which correspond to the two out of three bands of the semen from the vagina, which provides additional evidence against him;**

(students could label the bands and explain using the labeled terms)

[Total: /15 marks]

QUESTION TWO (15 MARKS)

Children with severe combined immunodeficiency disorder (SCID) cannot produce the many types of white blood cells that fight infections. This is because they do not have the functional gene to make the enzyme, adenosine deaminase (ADA).

Researchers are currently exploring the possibility of using stem cells in therapy for SCID. Some children with SCID have been treated with stem cells. Stem cells can divide and develop into any type of blood cell, including white blood cells.

The treatment used with the children is described in **Fig. 2.1**.

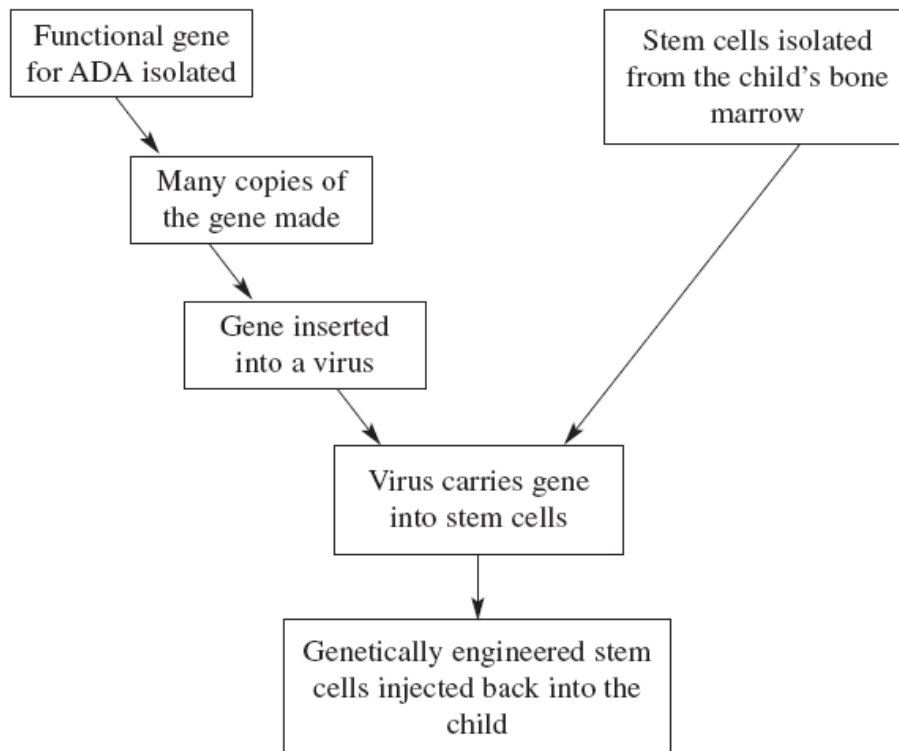


Fig. 2.1

- (a) Using the information given, state **two** reasons why stem cells were used in this treatment.

Capable of dividing for long periods to generate replacements for cells that are unable to produce ADA;

Unspecialised and can differentiate to specialized cell types such as white blood cells to fight infection;

The stem cells used belongs to the child and there will be no / little risk of rejection / triggering immune response; [any 2]

[2]

- (b) Suggest the type and the potency of stem cells that were used in this treatment.

Type Adult / hematopoietic / blood stem cells;

Potency Multipotent;

[2]

A child was treated with genetically engineered stem cells. **Fig. 2.2** shows the number of functioning white blood cells in the child during the year following treatment. Children who do not suffer from SCID have between 5000 and 8000 white blood cells per mm^3 of blood.

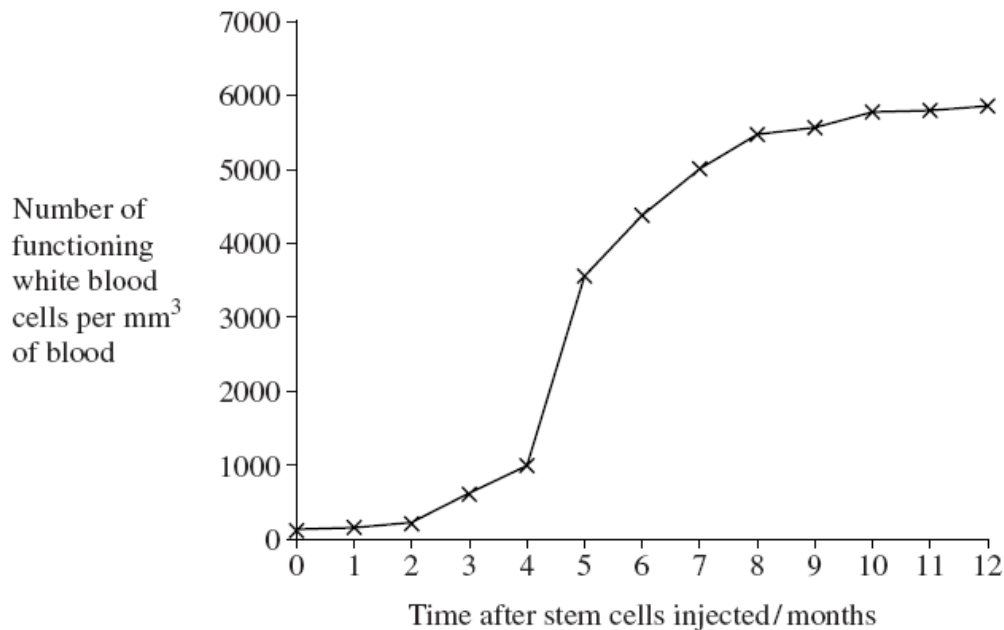


Fig. 2.2

(c) Describe the results shown in **Fig. 2.2**.

From 0 to 4 months of stem cells injection, the number of functioning white blood cells per mm^3 increases slowly from 100 to 1000;

From 4 to 8 months of stem cells injection, the number of functioning white blood cells per mm^3 increases exponentially from 1000 to 5500;

From 8 to 12 months of stem cells injection, the number of functioning white blood cells per mm^3 plateau / level off at 5800;

[3]

(d) Explain why a person treated by gene therapy may still have children who suffer from SCID.

Therapeutic gene inserted into somatic cells /

Therapeutic gene not inserted into gametes / germ cells;

Defective gene is inherited;

[2]

SCID is an inherited autosomal recessive disorder. Therefore it is easier to treat SCID by gene therapy.

- (e) Suggest why it is easier to perform gene therapy when a mutant allele is recessive instead of dominant.

When mutant allele is recessive, addition of functional / normal dominant allele by gene therapy will mask the effect of recessive allele /

If mutant allele is dominant, it has to be removed / repaired / inactivated to block production of defective gene product;

[1]

Although the use of gene therapy for treatment of genetic disorders in humans seems promising, there are still many concerns to be addressed.

- (f) Explain the factors that may keep gene therapy from becoming an effective treatment for genetic disorders.

Difficult to ensure that therapeutic gene is integrated into genome of host cells;

Incorrect insertion of therapeutic gene into genome of host cell results in cancer, as the insertion is random;

Immune response may be triggered resulting in rejection because a foreign virus vector is introduced;

Difficult to control expression of normal functional gene to give a fully functional protein;

[3]

- (g) Discuss **one** ethical and **one** social concern regarding the use of gene therapy.

Ethical: Gene therapy may be viewed as tampering with nature and going against the natural way of life;

Social: There may be genetic discrimination in the society as the issue of defining what is a disability/disorder/disease is unclear;

The poor will be discriminated against because they cannot afford the expensive treatment of gene therapy;

[2]

[Total: /15 marks]

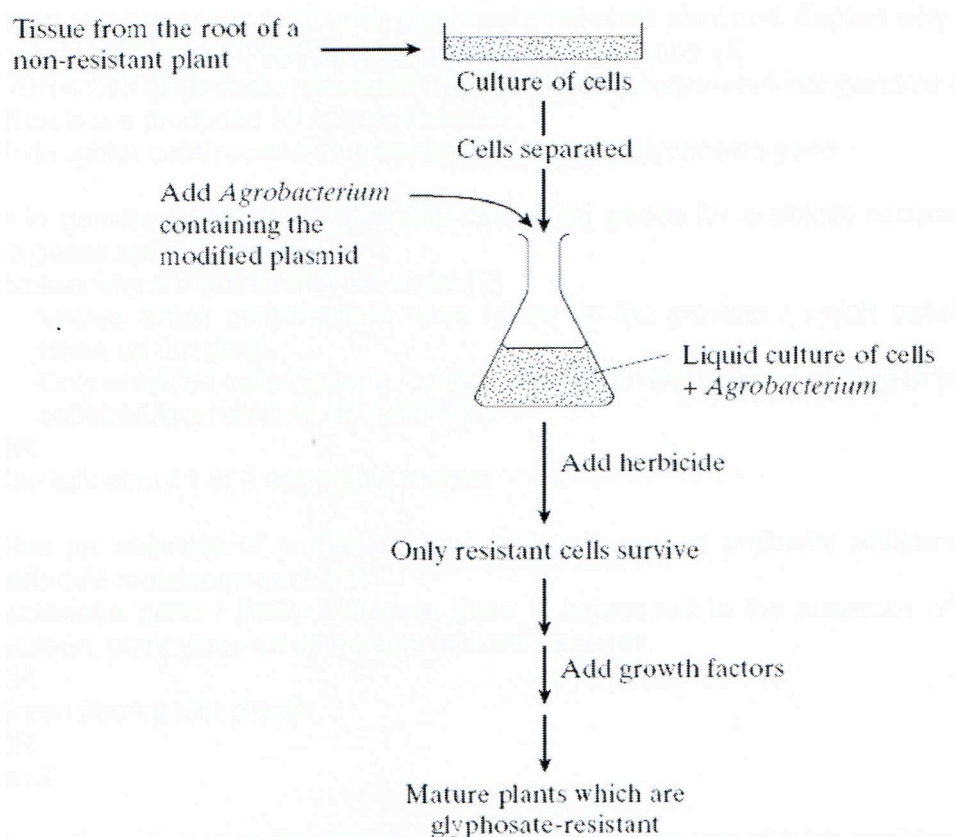
QUESTION THREE(10 MARKS)

Read the following passage.

Genetic engineering is a technique for improving crop plants. It is a process by which a single new gene can be added to the genes already present in the plants. One way of doing this is to use a natural genetic engineer, a soil microorganism called *Agrobacterium*. This bacterium possesses a plasmid which can be modified in the laboratory so that it becomes the carrier of new genetic information.

For example, a gene coding for the ability to break down the herbicide glyphosate can be introduced with the assistance of the enzymes restriction endonuclease and ligase. Once inside the crop plant, the gene for glyphosate breakdown makes the plant resistant to the effects of this herbicide. All surrounding weeds are destroyed when sprayed with glyphosate.

The flow-chart shows how the plasmid carrying this gene can be used to produce glyphosate-resistant plants.



Use the information from the passage and your own knowledge to answer the following questions.

- (a) One mature plant grows from each glyphosate-resistant plant cell. Explain why all the cells of the mature plant contain the gene for glyphosate-resistance. [1]

The cells are genetically identical / clone of the glyphosate cells due to mitotic division of the cell.

- (b) Outline the technique of tissue culture needed after obtaining matured plants from the transformed cells which are resistant to the herbicide. [3]

- The explant is first sterilized using sodium hypochlorite

The explant is then transferred to an aseptic culture medium that contains nutrients, growth factors/hormones.

- The cells of the explant will divide by mitosis to form a callus

The callus can be subcultured into many calli in different culture

- The ratio of plant hormones will result in the plant tissues developing into specialized tissues.

e.g. of a ratio leading to either root inducing or shoot inducing

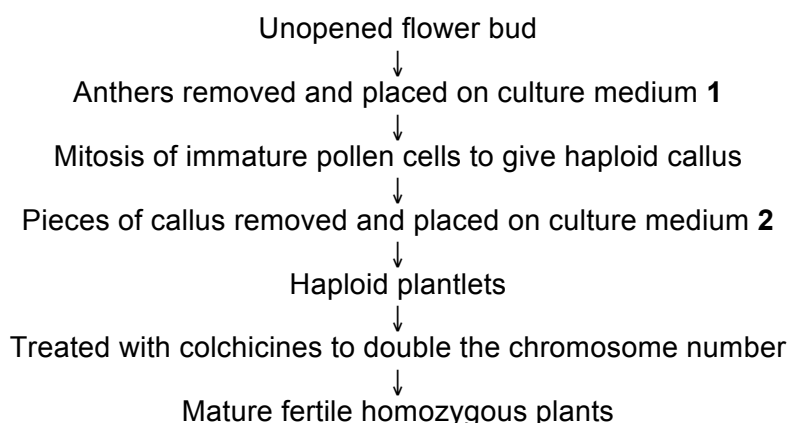
When a plantlet is formed, it can be transferred to soil to grow into a whole plant.

- © Suggest and explain **one** possible benefit and problem associated with the use of herbicides together with genetically modified, herbicide-resistant crop plants. [1]

- The creation of superweeds with horizontal gene transfer
- The production of toxins by the gene that can cause allergy in human beings
[Any 1 point]

Another technique required in plant biotechnology is plant tissue culture. Various organs or tissues of a plant may be used in tissue culture to produce a large number of genetically identical plants.

In the process of anther culture, haploid plantlets are produced from immature anthers, from which mature fertile homozygous plants are obtained. The process is shown diagrammatically in the figure below.



- (d) With reference to the figure,

- (i) explain what is meant by haploid callus. [2]

- The cells are genetically identical / undifferentiated glyphosate cells
- With 1 set of chromosomes

- (ii) State one precaution that would be taken when transferring material onto culture medium **1** or **2**. [1]

Ensure that the environment for the transfer is aseptic to prevent contamination.

- (iii) Suggest why variation may be generated in plantlets derived from the same callus. [1]

It can be due to chromosomal mutations

- (iv) Suggest how colchicine could bring about the doubling of chromosome number in haploid cells. [1]

It may prevent the sister chromatids from separating during anaphase/

Cause non-disjunction

[Total: /10 marks]

PLANNING QUESTION

QUESTION FOUR (12 MARKS)

You are required to plan, but **not** carry out, an investigation into the effect of XXX

Your planning must be based on the assumption that you have been provided with the following equipment and material which you must use:

- A large piece of beetroot tissue with the skin removed
- Sharp knife
- White tile
- Ruler
- Thermostatically controlled water bath
- Test-tubes
- Stopwatch
- Distilled water
- Colorimeter
- A variety of different-sized beakers, measuring cylinders and syringes for measuring volume

Your plan should have a clear and helpful structure to include:

- a description of the method used including the scientific reasoning behind the method,
- an explanation of the dependent and independent variables involved,
- relevant, clearly labelled diagrams,
- how you will record your results and ensure they are as accurate and reliable as possible,
- proposed layout of results tables and graphs with clear headings and labels,
- the correct use of technical and scientific terms,
- relevant risks and precautions taken.

Mark scheme

INTRODUCTION/Theoretical Background (2 marks)

1. Explanation of theory to support practical procedure
 - Define transformation as uptake of foreign DNA (pGLO plasmid) from the external environment by *E. coli* which expresses the genetic information encoded by the plasmid.
 - antibiotic resistance gene on plasmid will confer ampicillin resistance to the bacterial cells → able to survive when exposed to ampicillin to form colonies
 - arabinose in the agar medium will allow RNA polymerase to bind to promoter → allows for the transcription and translation of GFP gene → green colonies under UV light
2. Scientific reasoning behind the method
 - Ca^{2+} ions in transformation solution neutralize the negative charges of the plasmid DNA to facilitate its entry into the bacteria
 - Heat shock methods create temporary pores in bacterial cell membrane to facilitate the entry of plasmid

PROCEDURE/METHOD (Total number of points = 11 [3 – 9 & 11]; max is **6 marks**)

3. Preparation of different nutrient agar plates
(table/ any equivalent to show labels and composition of each plate)
4. Chemical-heat shock method for transformation
 - a) appropriate volume of competent cells + plasmid solution / sterile water + transformation solution in a microfuge tube
 - b) ice → 42°C, 50s heat shock → ice again
 - c) Method of achieving and maintaining desired temperature

Note that it is a **brief heat shock at 42°C for 50s** and there is **ice before and after** the step. Many students specify an incubation period of 10 – 30 minutes at room temperature to 40 °C instead which is for transduction.

5. a) Method of plating
Removal of suitable amount of bacteria → appropriate plate → spreader/ inoculating loop/ beads (spreading)

b) Incubation of bacteria for growth
(incubator oven, 37°C, overnight)
6. Use of control plate (any one)
(What – Grow the bacterial cells treated with sterile water instead of pGLO plasmid solution on LB agar plates
Why – to show that bacterial cells are still alive after going through the transformation procedure/ to compare with those cells grown on LB/Amp agar plates to show that ampicillin in the agar plate is effective in killing bacteria cells with no resistance
What – grow bacterial cells treated with sterile water on LB/Amp agar plates
Why – To compare with pGLO treated cells grown on LB/Amp agar plates to show that ampicillin resistance is indeed conferred by the pGLO plasmid
What – Grow pGLO treated bacterial cells on LB/Amp agar plate
Why – To compare with LB/Amp/Arabinose agar plate to show that Arabinose is causing fluorescence in pGLO treated bacterial cells)

7. Use of sterile equipment to prevent cross contamination between plates
(Use different sterile glass beads/ spreader/ inoculating loop for spreading the cells evenly across plate and new micropipette tips for every transfer of volume)
8. Repeat twice with replicates
(Idea of consistency → 6 sets of data)
9. Method of determining successful transformation
(White colonies / Green colonies and **count** the number of colonies that grow on nutrient plates)

USE OF DATA TO REACH A CONCLUSION

10. Labeled table for data collection (**1 mark**)
(Independent variable in leftmost column, headings: Name and type of agar plate, Observations: description of bacterial growth & (average) number of colonies; Experiment and replicates)
11. Processing of data
(Calculate **average** number of bacterial colonies)
12. Graph to represent processed data (**1 mark**)
(Plot **average** number of colonies against Name and type of agar plate – labels for x-axis, y-axis)

SAFETY ISSUE (1 mark)

13. Safety in handling of bacteria
(e.g. heat sterilization of bacterial plates at 121°C/ autoclaving before disposal / wear gloves when handling bacteria)
14. The use of correct technical and scientific terms (**1 mark**)
Scientific terms such as restriction enzymes, recombinant and re-annealed vector, replica plating are **not correct** terms to be used here as there is no need to cut the pGLO plasmid provided since it already has the GFP gene (GOI) inserted into it.

FREE-RESPONSE QUESTION (20 MARKS)

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labeled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections (a), (b) etc., as indicated in the question.

QUESTION FIVE

- (a) Describe PCR and compare and contrast with replication process. [6]
- (b) Outline the process of gel electrophoresis and explain how it is able to separate fragments of DNA. [9]
- (c) State the uses of RFLP analysis. [5]

Answers

- (a) Describe PCR and compare and contrast with replication process. [6]

1st step: One to several minutes at 94-96 degrees C, during which the DNA is denatured into single strands by breaking of hydrogen bonds.

2nd step: One to several minutes at 50-65 degrees C, during which the primers hybridize or "anneal" (by way of hydrogen bonds) to their complementary sequences on either side of the target sequence.

3rd step: One to several minutes at 72 degrees C, during which the Taq polymerase binds and extends a complementary DNA strand from each primer, using the single stranded DNA as a template.

	Replication	PCR
primer	RNA primers	DNA primers
enzyme	DNA polymerase III & I	Taq DNA polymerase
proofreading	yes	no
Temperature dependent	no	yes

(b)

Outline the process of gel electrophoresis and explain how it is able to separate fragments of DNA.

[9]

Agarose gel electrophoresis is a method for separating and visualizing DNA fragments produced by restriction digestion of DNA.

The fragments are separated by charge and size by forcing them to move through a agarose gel matrix which is subjected to an electric field.

Principles of agarose gel electrophoresis

- Biological molecules such as DNA, RNA and proteins are electrically charged particles at a given pH.
 - DNA is negatively charged due to the phosphate groups of the sugar-phosphate backbone.
 - If the sample contains DNA fragments of different sizes, they can be separated based on their sizes when they are subjected to an electric field.
 - Smaller fragments will move faster than the larger fragments.

Step 1: Casting of the agarose gel

- Agarose is a polysaccharide derived from seaweed that forms a gel when dissolved in an aqueous solution.
- Agarose solution will be poured into casting tray and allowed to solidify.
- With the help of the comb, small indentations called wells are formed at one end of the gel, where DNA samples can be loaded.
- Each well corresponds to one lane.

Step 2: Loading of sample

- DNA samples are first mixed with a loading dye which increases its density and helps the DNA to sink to the bottom of the well.
- Loading dye also helps to monitor the progress of the electrophoresis as DNA is invisible.
- Markers are usually run in the 1st lane. Markers are mixtures of DNA fragments of known sizes. By comparing the markers' fragment against the unknown fragments, we can estimate the sizes of the unknown fragments.
- Samples are loaded into wells with the help of a micropipette.

Step 3: Running of gel

- Current is switched on.
- Negatively charged DNA fragments will migrate out of the well and are attracted to the positive electrode.
- Separation of the fragments will be based on their sizes and can be seen as discrete bands on the gel.

Step 4: Visualization

- Before the band reaches the end of the gel, current will be switched off.
- Gel is then stained with DNA binding dye (ethidium bromide which is carcinogenic).
- Separated DNA fragments can be seen clearly under the UV light.

(c) State the uses of RFLP analysis.

[3]

- **Genomic mapping in terms of linkage mapping**
- **Disease detection, e.g sickle cell anaemia**
- **DNA fingerprinting**